

Student Stress Predictor using Machine Learning

PROJECT REPORT

Motivation

The motivation behind the creation of the student stress predictor using machine learning techniques originates from the rising levels of academic stress and competition faced by students in the current fast-paced academic world. Students face stress in their academic life due to various reasons like academic workload, academic deadlines, examinations, and academic goals.

These problems may not be recognized in the early stages until they affect the students' performance. The use of machine learning techniques in the creation of the student stress predictor can help in the early identification of the key causes of stress in students.

2. "Stress Prediction of Students using Machine Learning" by D. Sharma, N. Kapoor, and S. S. Kang:

Approach: Applied Random Forest, J48, and Naive Bayes algorithms on the PSS-based questionnaire data from over 200 students across various levels of academia.

Strength: A high accuracy of 94.73% was achieved using the Random Forest algorithm and used a wide range of performance metrics to validate the results of the predictive model.

Limitation: The work was limited by the low number of participants from a single university and the subjective nature of the self-reporting measures rather than objective physiological measures of stress.

Literature Review

For this project we studied two research papers :

1. "Analyzing Perceived Psychological and Social Stress of University Students: A Machine Learning Approach" by J. Ratul, M. M. Nishat, F. Faisal, S. Sultana, A. Ahmed, and M. A. Al Mamun:

Approach: Eight supervised models with PCA/Chi-squared reduction and Genetic Algorithm optimization on the PSS were used on survey data of 444 students.

Strength: A high predictive accuracy of 80.5% using the MLP algorithm was reported and the importance of feature reduction on precision was seen.

Limitation: The authors used convenience sampling and self-reporting, which is biased.

Dataset Description

This dataset was collected in order to analyze and predict the stress levels of students using machine learning algorithms. This dataset includes a variety of factors affecting student stress levels. This dataset contains 168 individual records collected using a survey. There are 9 predictor variables (features) and 1 target variable in this dataset. The data was collected mainly in March 2026.

The features of the dataset include various attributes of student lifestyle and academic behavior, such as age, gender, study hours, sleep hours, social media use, academic pressure, physical activities, and stress level. The target variable of this dataset is the stress level of the students.

#	Feature Name	Data Type	Values / Categories	Encoding / Processing
1	Age	Numeric	Actual numbers	None
2	Gender	Categorical	Female, Male	Label Encoding: Female → 0 Male → 1
3	Year of Study	Ordinal	0, 1, 2, 3, 4	None
4	Average Sleep Hours	Numeric	5, 6, 7	Text to Numeric Conversion: 5-6 hrs → 5 6-7 hrs → 6 >7 hrs → 7
5	Average Study Hours	Numeric	2, 4, 6	Text to Numeric Conversion: 2-4 hrs → 2 4-6 hrs → 4 >6 hrs → 6
6	Social Media Usage	Ordinal	0, 1, 2, 3	None (Already numeric)
7	Academic Pressure	Ordinal	1, 2, 3, 4, 5	None
8	Physical Activity	Categorical	Low, Medium, High	Label Encoding: Low → 0 Medium → 1 High → 2
9	Stress Level	Categorical	Low, Moderate, High	Target Variable

Data Preprocessing: The data set was preprocessed by removing inconsistencies and handling missing values appropriately, followed by converting all variables into numerical form. The features of the dataset were verified to be free of missing information.

1. Feature Name Standardization: The column names have been standardized to lowercase with underscores for better library compatibility. For example, "Year of Study" is now "year_of_study".

2. Handling Missing Values: The original data set had missing values in the column "Physical activity per week," which had 36 missing values, representing 21% of the total. The missing values have been handled by Imputation that is the missing values have been imputed based on the mode if the distribution.

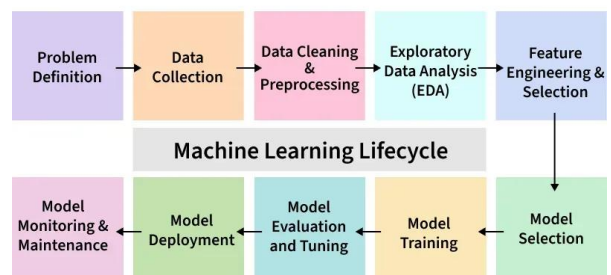
3. Feature Encoding and Transformation –

Original Feature	Transformation Applied	Encoding Logic
Gender	Label Encoding	0: Female, 1: Male, 2: Prefer not to say
Year of Study	Ordinal Mapping	0: 1st Year, 1: 2nd Year, 2: 3rd Year, 3: 4th Year, 4: Postgraduate
Average Sleep Hours	Bin Reduction	Values grouped and represented by lower bound: 5, 6, or 7 hours.
Average Study Hours	Bin Reduction	Values grouped and represented by lower bound: 2, 4, or 6 hours.
Social Media Usage	Ordinal Encoding	Ranges 0-3 representing duration (Very Low to High).
Physical Activity	Ordinal Encoding	Ranges 0-3 representing activity levels.
Stress Level (Target)	Ordinal Encoding	0: Low, 1: Moderate, 2: High

Exploratory Data Analysis (EDA): EDA was done in order to understand the pattern within the data set:

- Histograms were used for finding the distribution of numerical attributes
- Countplots were used for visualizing the distribution of categorical attributes
- Boxplots were used for understanding the outliers within the attributes such as sleep hours, study hours, etc., at varying levels of stress
- Some of the insights found from analysis include:
 1. Students who had lesser sleep hours had more stress levels
 2. Study hours beyond a certain level cause stress rather than helping in performing well
 3. Academic pressure acts as a major factor for stress

Methodology



Models used:

Logistic Regression was used as a baseline model due to its simplicity and interpretability in classification tasks.

Decision Tree model was applied to capture non-linear relationships between features such as academic pressure, sleep, and study hours.

Additionally, a **Random Forest** model, which is an ensemble of multiple decision trees, was used to improve prediction accuracy and reduce overfitting. By using multiple models, we ensured a comprehensive evaluation and selected the most effective model for stress prediction.

Evaluation Metrics:

Accuracy score was used to measure the overall correctness of the model's predictions.

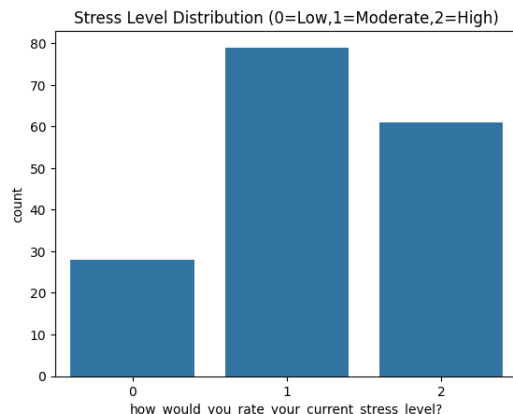
A **confusion matrix** was used to visualize how well the model classified different stress levels by showing correct and incorrect predictions.

Additionally, a **classification report** was generated, which includes precision, recall, and F1-score. These metrics together provide a detailed understanding of the model's performance.

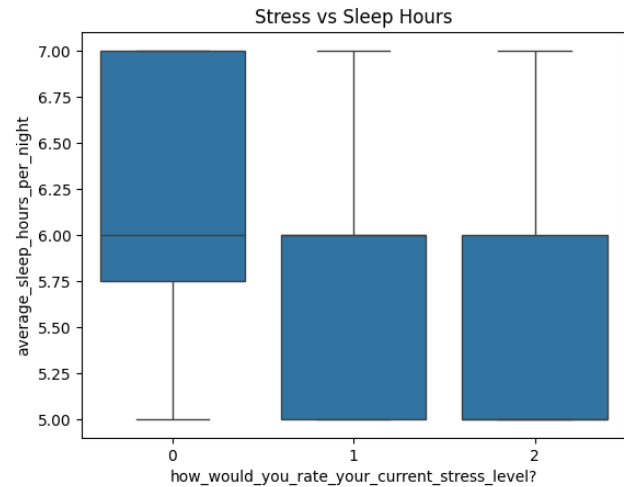
Analysis and Results

Various visualizations were created to understand patterns in the data:

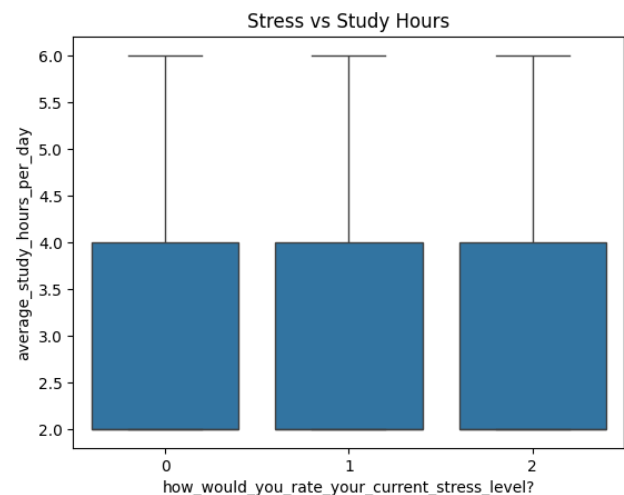
- **Stress Level Distribution:** Most students fall under moderate to high stress levels



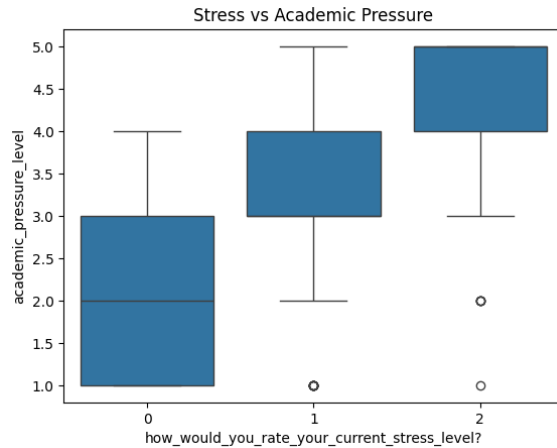
- **Sleep vs Stress:** Students with less sleep tend to have higher stress



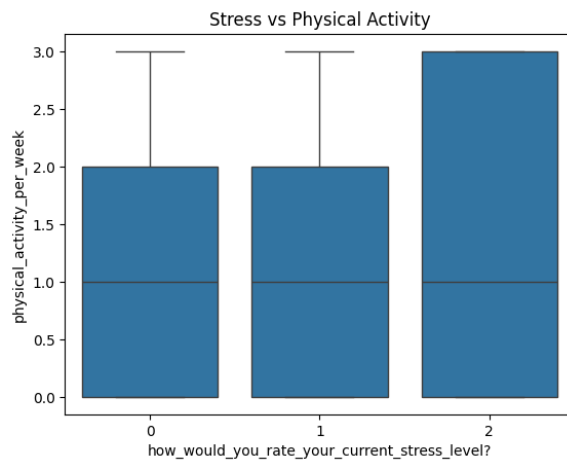
- **Study Hours vs Stress:** Excessive study hours are associated with increased stress



- **Academic Pressure vs Stress:** Academic pressure shows a strong positive correlation with stress



- Physical Activity vs Stress: Higher physical activity is linked with lower stress levels



Correlation Analysis:

A correlation heatmap was used to analyze relationships between variables

Observations:

- Academic Pressure has the strongest positive correlation with stress
- Sleep Hours has a negative correlation with stress
- Physical Activity also shows a negative relationship

- Study hours have a moderate positive effect

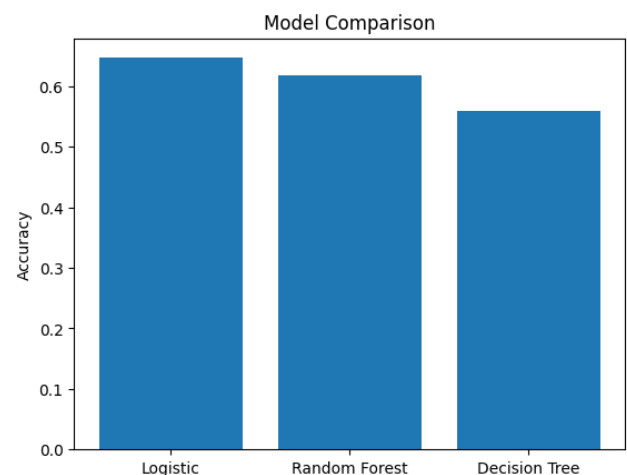


Results:

Logistic Regression Accuracy: 0.647%

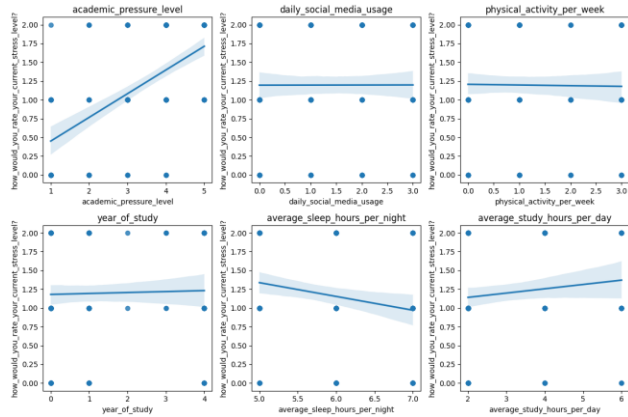
Decision Tree Accuracy: 0.558%

Random Forest Accuracy: 0.617 %



Among all models, **Logistic Regression achieved the highest accuracy** because the relationships between features like academic pressure, sleep, and study hours with stress levels are mostly linear. This model is well-suited for such data and can capture these patterns effectively. It is also

simpler and less prone to overfitting compared to complex models like Decision Trees, especially with smaller datasets. As a result, it generalizes better on unseen data, leading to higher accuracy.



Conclusion

The following insight were concluded:

1. Academic pressure is the most significant factor affecting stress
2. Sleep and physical activity play a crucial role in stress reduction
3. High study hours without balance increase stress levels
4. The model successfully predicts student stress levels with good accuracy

In this project, machine learning has been successfully utilized for predicting stress among students based on their academic and lifestyle features. After taking into consideration several features like study hours, sleep, and academic stress, it was found that all these models were able to classify the stress levels of students quite efficiently.

Out of all the models used in this project, Logistic Regression produced the best results in terms of accuracy. This shows that the correlation between the input features and stress level has a linear correlation, which is why Logistic Regression performed so well. In addition to that, it has been observed that sleep deprivation, study, and academic stress are some important factors that lead to higher stress.

Future Work

Improvements can be made in the present model by utilizing an extensive dataset for improving generalization. Other attributes like psychological elements, social behavior, lifestyle, and participation in extra-curricular activities could be added in order to get a better understanding of stress.

Various techniques from the realm of machine learning, including ensemble algorithms and deep learning methods, can be used for enhancing predictive power. Furthermore, the whole system can be built on the basis of a mobile application that would enable the continuous monitoring of the stress of students and provide personalized advice or treatment plans accordingly.